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A novel career prediction method based on fuzzy model—Fuzzy clustering approach

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ABSTRACT

Effective career decision-making strategies can help individuals enhance their competitiveness in the workplace, and assist companies in hiring more suitable employees. As an auxiliary tool for career decision-making, career prediction helps individuals to understand the professional world, assess career risks, broaden their career horizons, boost their career confidence, and ultimately make scientific career decisions. This study applied a fuzzy model for career prediction as it has the advantage of dealing with uncertainties, and can approximate the actual system with arbitrary accuracy. It divided individuals' sample data into occupational membership and vocational aspiration for employment and unemployment, respectively. Each sample data point included an individual's scores on 20 vocational interest items, 6 RIASEC scaled scores and their actual RIASEC code. Based on Holland's vocational interest theory, a fuzzy model's input and output variables are individuals' vocational interests and personality traits, respectively. As each personality trait corresponds to certain suitable occupations, a fuzzy model's prediction output will suggest the most fitting jobs for individuals. Moreover, to avoid the 'fuzzy rule explosion' problem when dealing with numerous input variables, fuzzy clustering is applied in fuzzy space partitions. As fuzzy clustering can automatically determine the number of fuzzy rules, it simplifies the fuzzy modelling process. When the fuzzy model-based career prediction method is compared with the profile and machine learning methods, the results show that the proposed career prediction fuzzy model has better prediction accuracy.

1. Introduction

Career decision-making and career prediction are complex sociological issues— involving psychology, sociology, and pedagogy, that play an important role in the development of individuals and organizations. Through career decision-making, college students can gain deeper insights into their interests, strengths, and values, and then, clarify their life goals. They will focus on acquiring professional knowledge, and reduce confusion or wastage of time during the university stage. They can also gain early insights into industry trends and market demands, and enhance their competitiveness in the job market by improving their professional skills, and making their practical experiences more targeted. Enterprise employees can improve their abilities of communication, teamwork, and problem-solving, and become more competitive in the workplace. Career decision-making helps individuals to recognize their missions and values, realize their self-worth in future work and life, enhance their sense of achievement, and create value for society. Furthermore, individuals can strengthen their adaptability, and establish extensive networks of contacts and resources to cope with career crises.

Given that, Frank Parson published 'Choosing a Vocation' in 1909, career decision-making theories have been developed over a period of more than 100 years. Among these theories, the ones with significant impacts are: Holland's vocational interest theory (Holland, 1959), Schein's career anchors theory (Schein and Van Maanen, 2013), Super's career decision theory (Super, 1980), Lofquist and Dawis's work adjustment theory (Dawis and Lofquist, 1978), Peterson's cognitive information processing theory (CIP) (Peterson et al., 1991), Ford and Lerner's developmental systems theory (DST) (Ford and Lerner, 1992), Lent, Brown and Hackett's social cognitive career theory (SCCT) (Lent et al., 1994), and Gati and Asher's personal competence and influence model (Gati and Asher, 2001). They all attempted to explain the process of people's career choices and development from different perspectives, and are still evolving and improving. In recent years, some new career

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decision-making methods—dual-process theory of career decision-making (Xu and Flores, 2023), online intervention (Pordelan et al., 2020), ambiguity aversion (Xu and Adams, 2020), meta-analysis (Udayar et al., 2020), longitudinal social network approach (Kim et al., 2023), etc. have been studied.

Career decision-making and career predictions are related to a certain extent, but their concerns and application fields are different. Career prediction is mainly based on data, trends, and personal characteristics to speculate and analyze future career development. It focuses on forecasting and evaluating possible future career paths, development opportunities, and challenges. It helps individuals and organizations to understand future career development possibilities by analyzing the career market, industry development trends, personal potential, etc. Career prediction does not directly determine actions, but provides references for future decisions. Conversely, career decision-making refers to the process, through which, individuals make career choices and plans based on current career information, interests, values, or market conditions. It emphasizes actions, and is a decision-making process of choosing an occupation from among multiple alternatives. Its purpose is helping individuals to make the choices that best suit their needs and goals, when faced with multiple career options. It focuses more on how to act in reality, and make the best decisions under current conditions. Career prediction is the basis for decision-making, as it provides directions and references for decision-making by predicting the future. Thus, accurate career predictions can significantly enhance individuals' career decision-making process.

Many factors—like motivation matters (Bargmann et al., 2022), math anxiety (Eidlin-Levy et al., 2023), and interest fit (Granillo-Velasquez et al., 2025), can effect career predictions. With the development of artificial intelligence (AI), research on the applications of AI technology in career fields have been gradually gaining attention (Bandhu et al., 2024; Bankins et al., 2024; Carson et al., 1999; Jassas and Mahmoud, 2022; Tan et al., 2024). The machine learning (ML) method—an important method for implementing AI through research data or statistical information, can analyze the intrinsic relationships of massive data, and establish systematic prediction models. ML consists of multiple branches, each having its own advantages or disadvantages, and it has achieved some successful applications in career prediction. Ster et al. predicted a career choice in family medicine using artificial neural networks (ANN) (Ster et al., 2015) which in turn, was applied in predicting career fulfillment among medical doctors in developing nations (Thomas et al., 2024), Gülten et al. developed a forecast model using ML algorithms for the Turkish Post Corporation (Gülten and Baraçlı, 2024), Zhang et al. applied deep learning in career planning and entrepreneurship of college students (Zhang and Wu, 2024), Zhu et al. developed a prediction model of career plateau for nurses using Logistic regression (LR) analysis (Zhu and Li, 2023). Additionally, Sun Yifan et al. explored how universities can accurately predict the post-graduation destinations of undergraduate students based on big data of student management information, and provided early employment services for undergraduates. They established a prediction model for graduates using the Lasso-Logistic algorithm and achieved an accuracy rate of over 70 % (Sun et al., 2019). Wang et al. utilized the eXtreme Gradient Boosting (XGBoost) algorithm to predict college students' career choices, and employed the Shapley Additive exPlanation (SHAP) method to explain the prediction results and influence of individual student attributes on the predictions (Wang et al., 2022). Guleria and Sood established a career prediction framework based on the White and Black Box models in ML, and used Explainable AI (XAI) to assist students in choosing suitable careers (Guleria and Sood, 2023). Yang and Chang adopted a data-driven approach, that used institutional data and social media information to predict students' career decisions (Yang and Chang, 2023). Nie et al. predicted the career decisions of over 4000 student groups using the Approach Cluster Centers based on XGBOOST (ACCBOX) model (Nie et al., 2020). Song et al. explored the application of ML methods, such as neural networks, k-nearest neighbors ($K-$NN), elastic net and random forests (RF) in career prediction and achieved better prediction results compared to the traditional profile method (Song et al., 2024). Fatima et al. established a career decision model based on MBTI's personality types using five ML methods, including the decision tree (DT), KNN, LP, RF and SVM (Fatima et al., 2022). Similarly, Hmood et al. established a career path selection model using five ML methods, including DT, bagging meta-estimator, gradient boosting, XGBoost and KNN (Al-Dossari et al., 2020). Hickey et al. used RF as an ensemble classifier to analyze gender differences in the career paths of over 2800 executives (Hickey et al., 2022).

Ever since the fuzzy sets theory was first proposed by Zadeh in 1965 (Zadeh, 1965), it has been widely applied in the fields of economics (Dechboon et al., 2021), management (Edjossan-Sossou et al., 2020), control (Fan and Li, 2024), identification (Ying and Lin, 2023), and so on. In contrast to traditional algebraic sets, where elements either belong or do not belong to algebraic sets, fuzzy sets apply varying degree of fuzzy membership to describe the degree to which an element belongs to certain fuzzy sets. This definition enhances fuzzy system's abilities for dealing with uncertainness. In 1970,Bellman and Zadeh published 'Decision-making in a fuzzy environment', which marked the beginning of research on fuzzy decisions (Bellman and Zadeh, 1970).Until now, fuzzy decision-making has been a hotspot research area in decision analysis (Hussain et al., 2021; Zhu and Ren, 2022), and been successfully applied in information systems outsourcing decisions (Nazari-Shirkouhi et al., 2017), credit scoring (Ignatius et al., 2018), multiobjective stochastic programming problems (Yano, 2016),outsourcing risks (Yazdani et al., 2021),sustainable circular supplier selections (Ecer and Torkayesh, 2022),assessing healthcare waste treatment (Rani et al., 2020),automating ergonomics risk assessments (Govindan and Li, 2023), and so on.

In the career decision-making field, some scholars have now published research articles on the fuzzy systems theory's applications, like Verma's three-dimensional model, that uses preference, fuzzy logic, and influence to recommend students' career paths (Verma et al., 2017), Ejegwa's Pythagorean fuzzy set in career placements (Ejegwa, 2019; Ejegwa et al., 2021), Yildiz's spherical fuzzy analytic hierarchy process in career management (Yildiz et al., 2020), Zhang's fuzzy-set qualitative comparative analysis in career paths aspiration (Zhang et al., 2024), Potwary's intuitionistic fuzzy sets in career determination (Potwary et al., 2024), Chen's fuzzy logic-based quantitative development model for job satisfaction (Cheng, 2024), Qamhieh's fuzzy intelligence of N-layered architecture (Qamhieh et al., 2020), Dasan's intuitionistic fuzzy sine distance measure (Dasan et al., 2024), and Khousa1's career development using fuzzy pairwise-constraints K-means (Khousa and Atif, 2018).

Fuzzy modelling—a widely used application of fuzzy systems theory for fuzzy models—can approximate the actual system with arbitrary accuracy (Wang and Mendel, 1992), and effectively handle real-world uncertainties and fuzziness, thereby making it suitable for scenarios with incomplete information or ambiguous definitions. Additionally, fuzzy models can handle complex and nonlinear problems more flexibly as its rules are usually expressed in natural language form, that is easy to understand and interpret. This study applies fuzzy models in career predictions, and divides the sample data into two distinct sets: 1) training data for determining fuzzy model parameters; 2) testing data for verifying the fuzzy model's accuracy. The results show that fuzzy models offer better prediction accuracy and significant guidance for the individual's career choices.

The rest of this paper is organized as follows: section 2—primary concepts of Holland's vocational interests theory, fuzzy sets, and fuzzy model; Section 3—methodology[i.e. the Takagi-Sugeno (T-S) fuzzy modelling process based on fuzzy clustering];Section 4—results of the fuzzy model, and how it compares with the profile and ML methods; Section 5—discussions; and section 6—conclusions and future work.

2. Primary concepts

2.1. Holland's vocational interests theory

Vocational interest is an important concept in the career development field, whose most widely used theory is Holland's theory of vocational personalities and work environments. By integrating concepts from personality psychology with extensive career counselling practices and research, Holland proposed his influential career interests theory (also known as the personality traits theory) in 1959. It points that individuals' career interests reflect their personality traits, which in turn, influence their career choices and career development, giving that, interests and personality traits are closely connected. Individuals with strong career interests tend to work and learn more actively. Based on different interests types, Holland categorized individuals into six personality types—Realistic, Investigative, Artistic, Social, Enterprising, and Conventional, collectively known as the RIASEC model. Characteristic of individuals with each of the six personality traits are described below.

- Realistic (R): They are skilled with tools and hands-on tasks, but lack strong social skills. They prefer concrete, practical work, making them well-suited for occupations, such as engineering and mechanics.
- Investigative (I): They are knowledgeable, analytical, enjoy abstract thinking, and thrive in independent and creative work, such as scientists and doctors.
- Artistic (A): They are creative, perfectionistic, seek to realize their individuality and personal values, and excel in roles, such as artists and musicians.
- Social (S): They possess excellent interpersonal communication, are eager to make a positive impact on society, and do well as educators, psychological counsellors, etc.
- Enterprising (E): They have strong leadership skills, are ambitious, and enjoy competition. They may pursue roles in sales (e.g. salespeople) or assume leadership positions (e.g. managers).
- Conventional (C): They respect authority and rules, but may lack creativity. They prefer structured environments, and do well as bank staff, data analysts.

2.2. Fuzzy sets

Fuzzy sets—the basic theory of fuzzy logic systems, are an extension of traditional algebraic sets, in which, elements can belong only to a certain set. Consequently, its membership degree can be either 0 or 1. In fuzzy sets, the membership degree of an element, represents the degree to which that element belongs to one fuzzy set, and can be any real number between 0 and 1.

Fuzzy sets $\boldsymbol{A}$ in the universe $\boldsymbol{X}$ is characterized by a membership degree $\mu_A(x)$ as (1).

$$A = \{(x, \mu_A(x))\,|x \in X\} \tag{1}$$

where $\mu_{\boldsymbol{A}}$:$\boldsymbol{X} \rightarrow [0,1]$, and $\mu_{\boldsymbol{A}}(x)$ represents the membership degree of the element x belongs to the fuzzy set $\boldsymbol{A}$. $\mu_{\boldsymbol{A}}(x)$ is calculated by membership function, and the most frequently used membership functions are triangular, Gaussian and trapezoid. Fig. 1 shows the triangular and trapezoid membership functions.

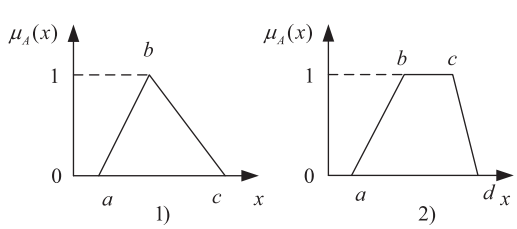

Fig. 1. 1) Triangular membership function, 2) Trapezoid membership function.

In Fig. 1.1, the triangular membership function is defined by 3 parameters (a,b,c) and the membership degree is calculated as (2).

$$\mu_A(x) = \begin{cases} \dfrac{b-x}{b-a}, a \le x < b \\ \dfrac{c-x}{c-a}, b \le x \le c \\ 0, else \end{cases} \tag{2}$$

In Fig. 1.2, the trapezoid membership function is defined by 4 parameters (a,b,c,d) and the membership degree is calculated as (3).

$$\mu_A(x) = \begin{cases} \dfrac{b-x}{b-a}, a \le x < b \\ 1, b \le x < c \\ \dfrac{d-x}{d-c}, c \le x \le d \\ 0, else \end{cases} \tag{3}$$

2.3. Fuzzy model

A fuzzy model is described by a series of if-then rules and the i-th fuzzy rule is defined as follow:

R^i : *if* x *is* A_i *then* $y_i = B_i$

Where, $i = 1,2, \ldots,c$, c is the number of fuzzy rules, $\boldsymbol{x}$ is the input vector composed by input variables(x_1,x_2,..,x_M), $\boldsymbol{A}_i$ is the primary part, $\boldsymbol{B}_i$ is the consequent part, y_i is the i-th fuzzy rule output. According to the forms of consequent parts, Mamdani and T-S are the commonly applied fuzzy models, whose consequent parts are crisp values and linear functions, respectively. The fuzzy model output can be calculated using the fuzzy inference system are shown in Fig. 2.

In Fig. 2, fuzzifer is the process that convert crisp input values into fuzzy variables and obtains the membership degrees of input variables within specified fuzzy sets. Fuzzy rules are generated by prior knowledge or expert experiences. Fuzzy engine typically applies min-max operator, and thereafter the defuzzifer process derives the final crisp fuzzy model output by centroid, weighted average, or center of sums methods.

Suppose one T-S fuzzy model has 2 inputs(x_1, x_2) and 1 output y, each input x_j defines 2 fuzzy setsA_j^1 and A_j^2. Then, 4 fuzzy rules will be generated as follows:

R^1 : *if* x_1 *is* A_1^1 and x_2 *is* A_2^1*then* $y_1 = p_1^1 x_1 + p_1^2 x_2 + p_1^0$

R^2 : *if* x_1 *is* A_1^2 and x_2 *is* A_2^1*then* $y_2 = p_2^1 x_1 + p_2^2 x_2 + p_2^0$

R^3 : *if* x_1 *is* A_1^2 and x_2 *is* A_2^1*then* $y_3 = p_3^1 x_1 + p_3^2 x_2 + p_3^0$

R^4 : *if* x_1 *is* A_2^2 and x_2 *is* A_2^2*then* $y_4 = p_4^1 x_1 + p_4^2 x_2 + p_4^0$

Define the membership degree for x_j in i-th rule is$\mu_i^j(x_j)$,which can be calculated by membership functions (2),(3) or other functions. Then, the fired membership degrees for i-th rule can be calculated by (4) under min fuzzy inference operator.

$$f_1 = \min(\mu_1^1(x_1), \mu_2^1(x_2)) \tag{4.1}$$

$$f_2 = \min(\mu_1^2(x_1), \mu_2^1(x_2)) \tag{4.2}$$

$$f_3 = \min(\mu_1^1(x_1), \mu_2^2(x_2)) \tag{4.3}$$

$$f_4 = \min(\mu_1^2(x_1), \mu_2^2(x_2)) \tag{4.4}$$

If the center of sums defuzzifer method is applied, the fuzzy model

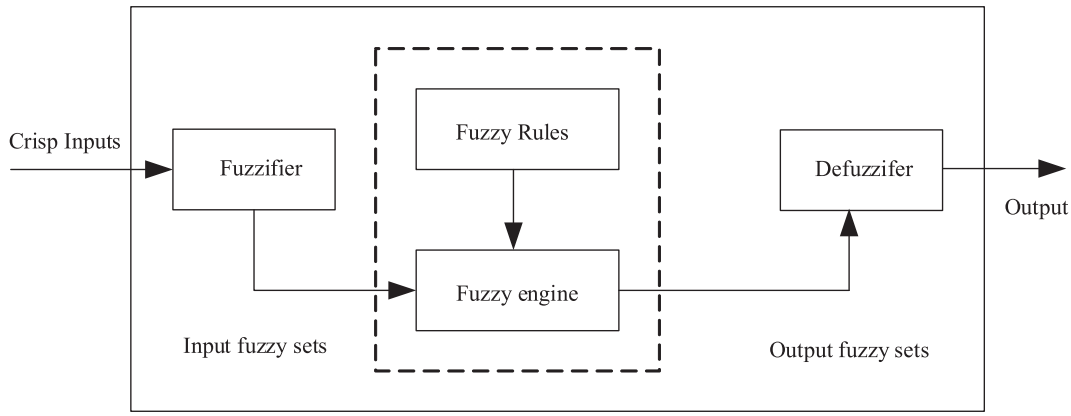

Fig. 2. Fuzzy inference system.

output y can be calculated by (5).

$$y = \frac{\sum_{i=1}^{4} f_i \times y_i}{\sum_{i=1}^{4} f_i} \tag{5}$$

3. Methodology

In practical implementations, if a fuzzy model involves multiple input variables or input variables define numerous fuzzy sets, it may lead to the problem of rule explosion. In this case, fuzzy clustering is a proper method for fuzzy modelling. Fuzzy clustering is the process of distinguishing and classifying sample data according to certain requirements and rules. Fuzzy clustering is an unsupervised learning method, that groups sample data into clusters based on the degree of similarity between data points. As their classification criteria revolves around assigning data points to clusters based on membership values, no prior knowledge about cluster assignments is required. Fuzzy c-means (FCM) clustering is widely used in fuzzy modelling as it can automatically generate membership degrees of fuzzy rules, and be easily implemented. The detailed procedure for FCM is described in Appendix A.

The T-S fuzzy model's i-th rule can be represented as follow:

R^i : *if x is* A_1 *then*$y_i = p_i^1 x_1 + p_i^2 x_2 + \cdots p_i^M x_M + p_i^0$.

For fuzzy modelling by FCM clustering algorithm, the first step is to determine the number of fuzzy clusters. This study applied the Xie-Beni cluster validity criterion (Xie and Beni, 1991) to determine the clustering number c.

$$v_{XB} = \frac{\sum_{i=1}^{c} \sum_{k=1}^{N} u_{ik}^m \|\theta_i - x_k\|}{N \min_{i \neq j} \|\theta_i - \theta_j\|} \tag{6}$$

In(6), $\|.\|$is the Euclidean norm and $\boldsymbol{\theta}_i$ is the i-th cluster representative. The Xie-Beni's index is considered more reliable than other cluster validity criteria because it considers all the information in the partition, and adopts the concept of compactness-to-separation ratio: Its numerator is a compactness validity function that fits the objective function of the fuzzy clustering algorithm to reflect the compactness of clusters, and its denominator is a separation validity function that measures the separation status of clusters. The smaller the separation validity function value, the higher the probability of redundant cluster representatives among the existing representatives. The optimal number c is then chosen when the Xie-Beni's index reaches its minimum.

For N pairs of input-output training data ($\boldsymbol{x}_k$,y_k), the clustering(fuzzy rule) number is c, the membership degree of $\boldsymbol{x}_k(k = 1,2, \ldots,N)$ corresponding to i-th rule is u_{ik}, $\boldsymbol{x}_k$ is a vector represented as ($\boldsymbol{x}_{k1}, \boldsymbol{x}_{k2}, \ldots, \boldsymbol{x}_{kM}$), M is the dimension of $\boldsymbol{x}_k$. The premise parameters of T-S fuzzy model is the center $\boldsymbol{\theta}_i(i = 1,2, \ldots,c)$ of training data, which can be calculated by the FCM clustering algorithm, and generally, its consequent parameters $p_i^j(j = 1, 2, \cdots, M)$ can be calculated by least squares estimation (LSE).

By FCM clustering algorithm, fired membership degree $f_i(\boldsymbol{x}_k)$ is the degree of $\boldsymbol{x}_k$ to i-th clustering membership degree u_{ik}. Then, for input $\boldsymbol{x}_k$, the fuzzy model output $\widehat{y}(k)$ can be calculated by (7).

$$\begin{aligned}\widehat{y}(k) &= \frac{\sum_{i=1}^{c} f_i(x_k) \times y_i}{\sum_{i=1}^{c} f_i(x_k)} = \frac{\sum_{i=1}^{c} f_i(x_k) \times \left(p_i^1 x_{k1} + p_i^2 x_{k2} + \cdots p_i^M x_{kM} + p_i^0\right)}{\sum_{i=1}^{c} f_i(x_k)} \\ &= \sum_{i=1}^{c} w_i(x_k) \times \left(p_i^1 x_{k1} + p_i^2 x_{k2} + \cdots p_i^M x_{kM} + p_i^0\right)\end{aligned} \tag{7}$$

In (7),

$$w_i(x_k) = \frac{f_i(x_k)}{\sum_{i=1}^{c} f_i(x_k)} \tag{8}$$

LSE is a classical method for parameter estimation. It can estimate the optimal parameters of a model that minimize the deviations between the actual output and fuzzy model output as (9)

$$\min \sum_{k=1}^{N} (y_k - \widehat{y}_k)^2 \tag{9}$$

For N pairs of training data, the input and output relation equation can be represented by (10).

$$Y = \widehat{X}.P \tag{10}$$

In (10),$\boldsymbol{Y}$ is the actual output vector indicated as (11).

$$Y = [y_1, y_2, \cdots, y_N]^T \tag{11}$$

$\widehat{X}$is the vector composed by input $\boldsymbol{x}_k$ and its corresponding fuzzy membership degree u_{ik}(that is $f_i(\boldsymbol{x}_k)$), and$\widehat{x}_k$is shown as (12).

$$\widehat{x}_k = [w_1(x_k).x_{k1} \ \cdots \ w_1(x_k).x_{kM} \ w_1(x_k) \ \cdots \ w_c(x_k).x_{k1} \ \cdots \ w_c(x_k).x_{kM} \ w_c(x_k)] \tag{12}$$

$\boldsymbol{P}$ is the T-S fuzzy model's consequent parameters vector as (13).

$$P = \left[p_1^1 \cdots p_M^1 \, p_0^1 \cdots p_1^c \cdots p_M^c \, p_0^c\right]^T \tag{13}$$

The estimate of $\boldsymbol{P}$ is calculated by LSE as (14).

$$P^* = \left(\widehat{X}^T \widehat{X}\right)^{-1} \widehat{X}^T Y \tag{14}$$

4. Results

4.1. Sample data

The sample data are cited from (Song et al., 2024), which contains vocational interests scores and self-reported RIASEC scores. The vocational interests scores are generated by Personal Global Inventory(PGI)-Mini. Totally, 20 questions are to be prepared for the respondents, and these are listed in Appendix B. The respondents will indicate their preferences toward each question, and get the corresponding vocational interests scores. The scores are on range 1 to 7,where 1 means strongly dislike, 4 means neural and 7 means strongly like. Table 1 displays one respondent's scores for 20 vocational interests.

Respondents' self-reported RIASEC scores are collected by O*NET occupation interest database, that has become a fundamental source of occupation information. It can be accessed through the O*NET center's website. It provides detailed descriptions of nearly 1000 occupation types. Each occupation's description includes 300 characteristics, such as worker requirements, knowledge domains, essential and cross-functional skills, work values, Holland occupational codes, tasks, educational requirements, experience and so on. The skills search allows individuals to assess their own skills and match them with suitable occupations in the database. Employed and unemployed respondents reported their occupation RIASEC scores by entering their professions and dream jobs, respectively. The highest value of 6 self-reported RIASEC scores comprises respondents' high-point code. Table 2 shows some examples of occupation RIASEC scores, and their high-point codes.

In accordance with the sample data, the career prediction fuzzy model's input variables comprised participants' scores on 20 vocational interests and 6 RIASEC scale items. The 6 RIASEC scale scores are calculated by the 20 vocational interests scores as follows: Realistic is calculated as: (item 5 + item 15)/2. Investigative is calculated as: (item 6 + item 16)/2. Artistic is calculated as: (item 7 + item 17)/2. Social is calculated as: [2 × (item 8 + item 18) + (item 1 + item 11)]/3. Enterprising is calculated as: [2 × (item 2 + item 12) + (item 1 + item 11)]/3. Conventional is calculated as: [2 × (item 4 + item 14) + (item 3 + item 13)]/3.

Sample data are divided into: occupational membership (for employment) and vocational aspiration (for unemployment), owing to their composition. The career prediction fuzzy model's output variables comprise 6 self-reported RIASEC scores of employed participants' occupational membership, or unemployed participants' vocational aspirations. As previously mentioned, 20 vocational interests scores are generated by PGI-Mini. However, the ML methods in (Song et al., 2024) selected items 1–8 and 11–18 as their input variables. For comparison, this study's fuzzy model shares the same input variables with ML methods. Hence, it includes 22 input variables (16 vocational interests and 6 RIASEC scale scores) and 6 output variables (6 self-reported RIASEC scores). The training and testing data are generated by 300 pairs of random sample data. Tables 3 and 4 display one occupational membership input sample data, whereas Table 5 displays one occupational membership output sample data.

4.2. Prediction results

Two methods are used to demonstrate the career prediction fuzzy model' accuracy. Firstly, the prediction effects are indicated by comparing the fitting degree curves of actual (self-reported) and predicted RIASEC scores. In this study, the fitting degree is depicted by the root mean square error (RMSE)—the identification accuracy index, which is calculated as (15), where N is the number of sample data, y_k is the actual RIASEC score and $\hat{y}_k$ is the predicted RIASEC score.

$$RMSE = \sqrt{\sum_{k=1}^{N} (y_k - \hat{y}_k)^2 \Big/ N} \tag{15}$$

Fig. 3 shows the actual and predicted RIASEC scores curves for occupational membership. These curves are generated by 300 random pairs of sample data (only 100 are displayed). The blue and red lines represent actual and predicted RIASEC scores, respectively, as defined in Fig. 4.

The RMSE for occupational membership of the 6 RIASEC scores curves are 1.61,1.76,1.48,1.67,1.84 and 1.43 respectively.

Fig. 4 shows the actual and predicted RIASEC scores curves for vocational aspiration, generated by 300 random pairs of sample date (only 100 are displayed).

The RMSE for vocational aspiration of the 6 RIASEC scores curves are 1.68,1.86,1.81,1.75,1.81 and 1.40 respectively.

Secondly, three prediction accuracy evaluation methods are applied: (1) highpoint hit rates(including both overall and differential hit rates of each RIASEC occupational category),(2) Euclidean distance, and (3) profile correlations.

The hit rate for a career prediction model represents the proportion of instances where the model's predicted occupation matches the actual occupation, relative to the total number of data points in the sample. For example, if a sample's max RIASEC score is R, then its high-point code is also R. Similarly, if the predicted model's max RIASEC score is R, then the R category's hit count will be incremented by adding one. Table 6 shows 2 examples of the fuzzy model's hit operations. Example 1—the predicted and self-reported high-point codes are both E; so, the fuzzy model accurately hit the self-reported high-point code. Contrarily, in example 2, it did not hit the self-reported high-point code.

Euclidean distance is a fundamental geometric concept that measures the straight-line distance between two points in Euclidean space. In this study, it represents the distance between the self-reported and predicted RIASEC scores. Profile correlations usually measure the statistical association between two or more profiles (e.g. vectors or data arrays), each representing multiple variables or observations under varying conditions. Higher highpoint hit rates or smaller Euclidean distances indicate better prediction accuracy.

Table 7 presents a comparison of hit rates, Euclidean distance, and profile correlation among the fuzzy model, profile method, and ML method. The bolded values indicate the highest hit rate, highest profile correlation and lowest Euclidean distance.

Tables 8 and 9 show the high-point hit rates of the 6 RIASEC occupational categories for occupational membership and vocational aspiration, respectively. The base rates comprise the proportion of hit rates for each RIASEC occupational category to the total sample data. The absolute value of differential hit rates denote the proximity between the prediction method's hit and base rates, the bolded values denote the

Table 2
Examples of occupation RIASEC scores.

Occupation	R	I	A	S	E	C	High-point code
Chief Executives	1.33	2	2.67	3.67	7	5.33	E
Cytotechnologists	5	5.67	1.67	1.67	1.67	3.33	I
Surgical Assistants	4	3.33	1.33	5	1.67	6.33	C

Table 1
One respondent's scores for 20 vocational interests.

20 vocational interests scores																			
1	1	4	5	1	6	1	1	1	1	1	3	4	1	1	4	1	1	1	1

Table 3
16 vocational interests scores in input sample data.

16 vocational interests scores															
1	1	4	5	1	6	1	1	1	3	4	1	1	4	1	1

Table 4
6 RIASEC scale scores in input sample data.

6 RIASEC scale scores					
1	5	1	1	1.666667	3.333333

Table 5
6 self-reported RIASEC scores output sample data.

6 self-reported RIASEC scores					
7	4.67	1.33	1.33	1.33	4.33

highest differential hit rate among the fuzzy model, profile and ML methods. SDs refer to standard deviation, in Tables 8, 9, and 10. In accordance with hit rates, higher SDs indicate better prediction accuracy, but for differential hit rates, lower means and SDs indicate better prediction accuracy.

Table 10 shows the Euclidean distance of each RIASEC occupational category, the bolded values denote the lowest Euclidean distance among the fuzzy model, profile method, and ML method, with lower means and SDs indicating better prediction accuracy.

Table 11 shows the profile correlation of each RIASEC occupational category, with bolded values denoting the highest profile correlations among the fuzzy model, profile method, and ML method.

The reliability coefficients (Cronbach's alpha) for each of the six RIASEC occupational categories are 0.49, 0.51, 0.50, 0.50, 0.45, and 0.44, respectively, for occupational membership, and 0.45, 0.47, 0.58, 0.35, 0.48, and 0.54, respectively, for vocational aspiration.

5. Discussion

Traditional clustering analysis is a type of hard partition that strictly divides the objects into distinct categories, so that the boundaries between them are clear. However, in reality, most objects may belong to several categories, their boundaries are unclear, and it's a soft partition, that cannot be solved by hard partitions. Fuzzy clustering analysis provides a powerful analytical tool for such soft partitions. Fuzzy space partition is a critical step in the fuzzy modelling process, particularly after the fuzzy model inputs are confirmed. In general, as fuzzy space domains overlap, fuzzy clustering is a proper fuzzy space partition method for fuzzy modelling. In this study, the proposed career prediction fuzzy model contains 22 input variables, by fuzzy clustering, the number of fuzzy rules is the clustering number, it avoids the fuzzy rule explosion problem.

This study's career prediction mode is based on Holland's vocational interests theory. Interest is an important driving force for individuals' career choices, hence career predictions based on interests can help individuals to find careers that best match their individual personalities. The career prediction model can be built upon the relationship between career interests and Holland's personality types. With the predicted personality type, individuals can choose their proper occupations. For instance, investigative personality type individuals are ideal for working as scientists, engineers or data analysts, social personality type individuals as teachers, psychological counsellors or social workers, enterprising personality type individuals as managers, marketers, entrepreneurs, etc.

The career prediction fuzzy model's accuracy is displayed in two ways: 1) the fitting degree of self-reported and predicted RIASEC scores curves by the RMSE index. From Figs. 3 and 4,it can be seen that the fitting degree is good and RMSE is relatively low, and 2) the highpoint hit rates, Euclidean distance, and profile correlations. Table 7 compares individuals' RIASEC overall scores for occupational membership and vocational aspiration, which show that fuzzy model has the lowest Euclidean distance, highest highpoint hit rates, and profile correlations.

Tables 8 and 9 compare the differential high-point hit rates of each RIASEC occupational category for occupational membership and vocational aspiration. The results indicate that for each RIASEC occupational category, the fuzzy model's differential hit rate is smaller than that of the profile or ML methods, which means that the fuzzy model does not perform better than the profile or ML methods because its accuracy is closely related to the number of samples. A large amount of samples can

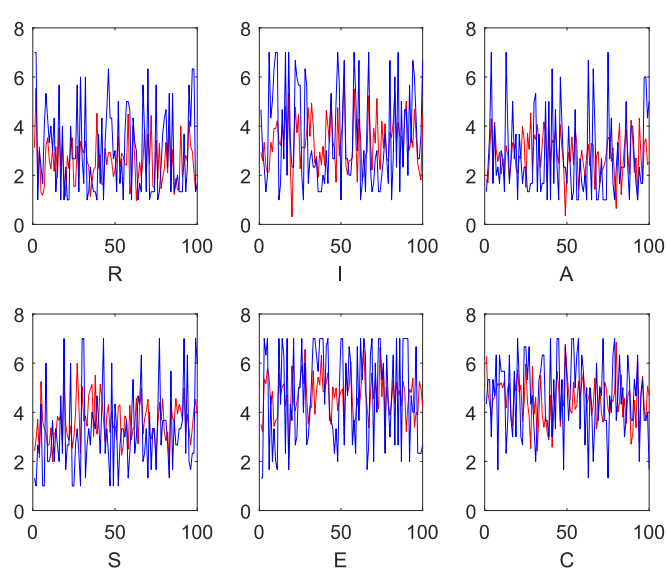

Fig. 3. The actual and predicted RIASEC scores curves for occupational membership.

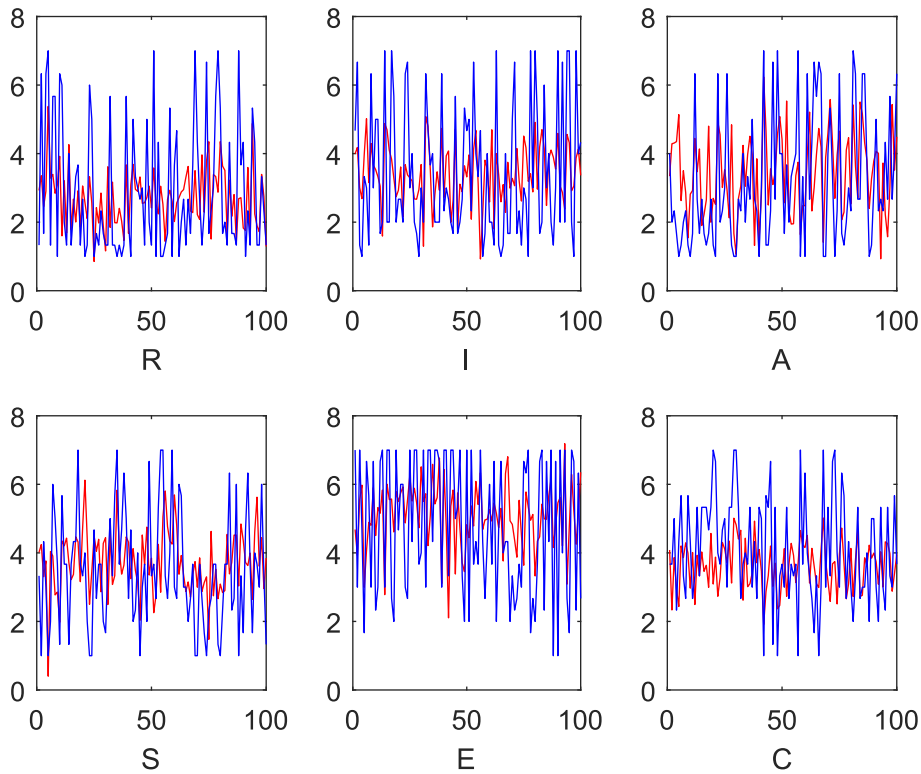

Fig. 4. The actual and predicted RIASEC scores curves for vocational aspiration.

Table 6
Examples of fuzzy model hit operation.

Example		Fuzzy model		Self-reported		Hit
		Category	High-point code	Category	High-point code	
1	R	1.59	E	1	E	Hit
	I	3.33		2.33		
	A	2.90		3.67		
	S	3.71		2.67		
	E	5.35		7		
	C	4.35		5.33		
2	R	3.11	C	7	R	No hit
	I	2.93		4.67		
	A	1.82		1.33		
	S	2.42		1.33		
	E	3.5		1.33		
	C	6.28		4.33		

improve the fuzzy model's accuracy. In terms of result, only 300 pairs of sample data are applied. For instance in Table 8, the R personality has a low base rate(=0.07, the training number is 21), that denotes a rare training sample data for it. Thus, the fuzzy model' accuracy is not the best. The career prediction model's accuracy can be improved by increasing the number of training data and making sure that the number of each RIASEC occupational category is uniformly distributed.

Table 10 compares the Euclidean distance of each RIASEC occupational category, with the fuzzy model having the lowest Euclidean distance. Table 11 compares the profile correlations of each RIASEC occupational category and in most cases, the fuzzy model has the highest profile correlations.

6. Conclusions and future work

Fuzzy model is described by the if-then fuzzy rules, and is easy to understand and explain. It is especially suitable for situations that lack precise mathematical models. This study explores the fuzzy model's application in career prediction, and uses it to propose a career prediction method in accordance with Holland's vocational interests theory.

Table 8
High-point hit rates of each RIASEC occupational category for occupational membership.

	Hit rate		Differential hit rate		
	Base rate	Fuzzy model	Profile method	ML model	Fuzzy model
R	0.07	0.01	0.11	−0.07	−0.06
I	0.19	0.09	0.09	−0.13	−0.1
A	0.04	0.00	0.39	−0.06	−0.04
S	0.13	0.09	0.04	−0.11	−0.04
E	0.41	0.37	0.03	0.44	−0.04
C	0.16	0.44	−0.06	0.04	0.28
Means	0.16	0.16	0.1	0.018	0
SDs	0.12	0.17	0.14	0.19	0.12

Table 7
The hit rate, Euclidean distance, and profile correlation results.

	Occupational membership			Vocational aspiration		
	Hit rate	Euclidean distance	Profile correlation	Hit rate	Euclidean distance	Profile correlation
Profile method	0.28	5.18	0.23	0.34	4.97	0.28
ML method	0.34	4.07	0.44	0.36	4.25	0.36
Fuzzy model	0.39	3.89	0.52	0.42	4.06	0.44

Table 9
High-point hit rates of each RIASEC occupational category for vocational aspiration.

	Hit rate		Differential hit rate		
	Base rate	Fuzzy model	Profile method	ML model	Fuzzy model
R	0.07	0.02	0.10	−0.04	−0.05
I	0.19	0.08	0.09	0.03	−0.11
A	0.23	0.17	0.19	0.12	−0.06
S	0.11	0.05	0.11	−0.08	−0.06
E	0.32	0.65	0.13	0.48	0.33
C	0.08	0.03	0.13	−0.04	−0.05
Means	0.16	0.16	0.125	0.078	0
SDs	0.08	0.22	0.03	0.19	0.14

Table 10
Euclidean distance of each RIASEC occupational category.

	Occupational membership			Vocational aspiration		
	Profile method	ML method	Fuzzy model	Profile method	ML method	Fuzzy model
R	5.79	5.13	1.29	5.80	5.12	1.33
I	5.07	4.63	1.45	5.00	4.56	1.55
A	4.52	4.96	1.24	4.76	4.01	1.49
S	5.29	4.65	1.38	5.06	4.52	1.44
E	5.00	3.43	1.58	4.81	3.83	1.50
C	5.49	3.63	1.15	5.08	4.36	1.14
Means	5.19	4.4	1.34	5.08	4.4	1.40
SDs	0.39	0.64	0.14	0.34	0.41	0.13

Table 11
Profile correlations of each RIASEC occupational category.

	Occupational membership			Vocational aspiration		
	Profile method	ML method	Fuzzy model	Profile method	ML method	Fuzzy model
R	0.04	0.02	0.41	0.02	−0.11	0.36
I	0.24	0.20	0.41	0.24	0.17	0.41
A	0.34	−0.14	0.41	0.31	0.47	0.47
S	0.16	0.11	0.39	0.23	0.12	0.26
E	0.32	0.75	0.39	0.38	0.63	0.38
C	0.18	0.72	0.31	0.28	0.39	0.48

The fuzzy model's inputs and outputs are individuals' interests and Holland's occupational codes (6 RIASEC scores), respectively. The predicted RIASEC scores will confirm individuals' personality traits, and thereafter suggest suitable occupations to them. As the career prediction fuzzy model has numerous inputs, its process involves applying the FCM clustering algorithm, that coincides with fuzziness data in the real world, and is more robust for noise data. It is directly applied in premise parameters identification for the T-S fuzzy model, which simplifies the fuzzy modelling process. The results show that the fuzzy model has better prediction accuracy.

The studied career prediction method only utilities individuals' interests as inputs for its fuzzy model, whereas in reality, other factors may affect individuals' personality traits. Future research must focus on exploring more affecting factors as inputs to improve the fuzzy model's prediction accuracy. Furthermore, the proposed career prediction method can provide auxiliary support for career decision-making, particularly, fuzzy career decision-making.

CRediT authorship contribution statement

Li Zhang: Writing – original draft. **Xueying Gu:** Conceptualization.

Declaration of Generative AI and AI-assisted technologies in the writing process

The authors did not use AI for any part of the work related to the manuscript submitted.

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Declaration of competing interest

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Appendix A. The FCM clustering algorithm procedure

For the sample data $\boldsymbol{x}_k(k = 1,2, \ldots,N)$, N is the number of sample data, cluster the data to c categories, the goal of FCM is to minimize the following objective function:

$$J_m(U,\theta) = \sum_{i=1}^{c}\sum_{k=1}^{N} u_{ik}^m d_{ik}^2 \tag{A-1}$$

where, u_{ik} is the degree to which data $\boldsymbol{x}_k$ belongs to cluster i, m is a weighting exponent and m > 1, and in general, $m = 2$, d_{ik} id the Euclidean distance of $\boldsymbol{x}_k$ to center $\boldsymbol{\theta}_i$.

1) . Set identification parameters: number of clustering c, maximum iteration number r_{max} and threshold ε, ε is a sufficiently small positive number. Initialize clustering center $\boldsymbol{\theta}^{(0)}$, set iteration count $r = 1$.
2) . Update the clustering degree u_{ik} by (A-2).

$$u_{ik}^r = \frac{1}{\sum_{j=1}^{c}\left(\frac{d_{ik}^r}{d_{jk}^r}\right)^{\frac{2}{m-1}}} \tag{A-2}$$

where, $d_{ik} = \|\theta_i - x_k\|$.

3) . Update the clustering center $\boldsymbol{\theta}_i$ by (A-3).

$$\theta_i^{r+1} = \frac{\sum_{k=1}^{N} \left(u_{ik}^r\right)^m x_k}{\sum_{k=1}^{N} \left(u_{ik}^r\right)^m} \tag{A-3}$$

4) . if$\left\|\theta^{(r+1)} - \theta^r\right\| \leq \varepsilon$or $r = r_{max}$, end the procedure; otherwise, set $\boldsymbol{\theta}^r = \boldsymbol{\theta}^{(r+1)}$, $r = r + 1$ and go to Step 2).

In this study, the convergence threshold ε is 1×10^{-5} and maximum number of iterations r_{max} is 100.

Appendix B. 20 questions to be prepared for respondents

question1	Seat patrons at a restaurant
question2	Oversee a hotel
question3	Prepare financial reports
question4	Oversee a data analysis group
question5	Install electrical wiring
question6	Categorize different types of wildlife
question7	Sculpt a statue
question8	Help children with learning problems
question9	Give lecture to large groups
question10	Drive a bus
question11	Interview people for a survey
question12	Manage an office
question13	Maintain office financial records
question14	Manage an electrical power station
question15	Oversee building construction
question16	Write a scientific article
question17	Paint a portrait
question18	Teach people to dance
question19	Study the effects of elections
question20	Carry and load containers

Data availability

The sample data is available at the website: https://osf.io/27rbh/files/osfstorage.

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